

Katherin Molina Granados

Computational Researcher — Autonomous Systems, Multi-Agent Systems, Computational Modelling

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Research Profile

Computational researcher with an interdisciplinary background in Information Technology, Business Analytics, and software engineering. Current research focuses on coordination, communication, and distributed decision-making in heterogeneous autonomous systems operating under uncertainty.

Develops computational frameworks combining Multi-Agent Systems, Agent-Based Modelling, distributed artificial intelligence, experimental design, and software engineering — examining how autonomous software agents, robotic platforms, and human supervisors share mission information, allocate tasks, coordinate resources, and adapt when operational conditions change.

Long-term research objective: generalizable frameworks for real-time mission communication, adaptive task execution, and collaborative autonomy in complex field environments.

Research Interests

Primary Research Area: Mission-Level Coordination for Heterogeneous Autonomous Systems

Core Research Areas: Multi-Agent Systems, Mission-Level Autonomy, Heterogeneous Autonomous Systems, Distributed Decision-Making, Multi-Robot Coordination, Human–Autonomy Teaming, Autonomous Cognitive Systems

Supporting Research Areas: Agent-Based Modelling, Distributed Artificial Intelligence, Agent Communication, Adaptive Task Allocation, Coalition Formation, Agent Governance and Oversight, Complex Adaptive Systems, Computational Simulation

Application Domains: Field Robotics, Environmental Monitoring, Precision Agriculture, Autonomous Mining Operations, Complex Industrial Systems

Education

Master	Pontificia Universidad Católica de Chile , Business Analytics	Chile
	- Graduated with Maximum Distinction; GPA 6.7/7.0 - Ranked 1st in Graduation Seminars I, II, and III; overall ranking 4th of 40 graduates - Research on the integration of machine learning, topological segmentation, survival analysis, explainable AI, and financial simulation	Jan 2024 – Mar 2026
	Diploma Pontificia Universidad Católica de Chile , Business Intelligence	Chile
	GPA 6.6/7.0 — business intelligence, data analytics, decision-support systems, data warehousing, organisational performance analysis	Jan 2024 – Dec 2024
Bachelor of Information Technology	Victoria University , Web & Mobile Application Development	Australia
	- GPA 6.13/7.0 — degree completed in Australia and delivered in English; software development, systems analysis, databases, object-oriented programming, computer networks, digital transformation - Capstone: designed, engineered, and deployed a full social network platform for Victoria University, covering the complete software engineering lifecycle from design through deployment	Jan 2020 – Dec 2023

Undergraduate (5 semesters) **Universidad Cooperativa de Colombia**, Telecommunications and Electronics Engineering

- Foundational studies in electronics, telecommunications, mathematics, and engineering systems (Algebra, Calculus, Probability & Statistics, Algorithm Design & Logical Problem Solving, Wave Physics, Circuits & Signal Propagation)

Colombia
Jan 2009 – Dec 2011

Certificates

Diploma in Business

AWS Data Analytics

Diploma in Project Management (Universidad EAN/Colombia)

Publications

Structural Coordination Capacity, Not Incentive Intensity, Governs Decentralized Heterogeneous Multi-Robot Missions: Evidence from the MOSAIC Agent-Based Model

July 2026

With Lorena Holguin, Software Engineer at Kahlo Group. Zenodo preprint.

Katherin Molina, Lorena Holguin

zenodo.org/records/21718799

MOSAIC v1.1.0 — Mission-Oriented Self-Organization through Auctions, Incentives, and Coalitions

Aug 2026

CoMSES Computational Model Library (peer-reviewed).

Katherin Molina

www.comses.net/codebases/2174fbc6-55cd-4f6e-abc7-9a79aa25ef81/releases/1.1.0

A Multi-Layer Analytical Framework for Customer Retention in Voluntary Health Insurance

Apr 2026

Zenodo preprint.

Katherin Molina

zenodo.org/records/19815868

Projects

AOOS — Autonomous Organization Operating System

Jan 2023 – present

Kahlo Group. Industrial research and technology development programme combining investigation of autonomous cognitive and multi-agent systems with the engineering of an enterprise software platform for autonomous organisations. The research dimension examines how heterogeneous autonomous software agents can communicate, coordinate tasks, share organisational knowledge, and make distributed decisions while remaining subject to explicit governance and human oversight; the product dimension translates these concepts into a platform for managing autonomous agents, organisational knowledge, permissions, workflows, policies, resources, and decision processes.

- Multi-agent orchestration, autonomous cognitive systems, and agent communication protocols
- Agent capability/role representation and organisational knowledge ontologies
- Workflow and task orchestration; policy and permission enforcement
- Agent memory and knowledge management
- Auditability, decision traceability, real-time observability, and explainable agent behaviour
- Human-in-the-loop supervision and human-autonomy teaming across heterogeneous agents and systems

Experience

Kahlo Group Pty Ltd, CPTO

Leads technology, analysis, and research-oriented initiatives involving organisational systems, software architecture, business processes, and applied artificial intelligence.

- Development of the AOOS research and technology programme
- Design of architectures for autonomous software agents and investigation of multi-agent coordination and governance mechanisms
- Translation of organisational requirements into technical architectures
- Development of structured workflows, information models, and decision systems; integration of research, engineering, and product-development practices
- Design of auditability, observability, and human-supervision mechanisms for autonomous systems

Sydney, Australia

Jan 2015 – present

11 years 9 months

KITBC Consulting, Consultant Data Analyst

Short-duration analytics engagement supporting operational improvement for large-scale automated mining operations.

- Analyzed operational information to identify performance bottlenecks and process inefficiencies
- Direct exposure to autonomous fleet operations, operational analytics, resource utilisation, and the interaction between autonomous technology and operational management

Feb 2025 – Feb 2025

1 month

Open Source Projects

MOSAIC — Agent-Based Multi-Robot Coordination

Independent computational research, 2026. Designed and evaluated an open, reproducible Agent-Based Model for decentralized coordination mechanisms in heterogeneous multi-robot missions, integrating task allocation, communication networks, auctions, incentives, reputation, coalition formation, and adaptive agent behaviour.

- Formalization of individual and cooperative mission tasks; representation of heterogeneous robot capabilities
- Design of market-based task-allocation mechanisms and coalition-formation procedures
- Implementation of communication and distributed-knowledge mechanisms
- Definition of reward, reputation, and adaptation processes; construction of automated verification procedures
- Development of a reproducible 690-run experimental programme and statistical evaluation of structural and behavioural coordination mechanisms

Multi-Layer Customer Retention Framework

Master's research project, 2025–2026. Developed a computational framework for customer-retention decision-making using real-world administrative records from a Colombian health insurer, integrating several analytical perspectives because the underlying problem could not be adequately represented by a single prediction model.

- Designed an end-to-end analytical research pipeline integrating topological segmentation, machine learning, and survival analysis
- Implemented explainability using SHAP; compared alternative predictive and survival models
- Evaluated behavioural, temporal, and financial dimensions of customer retention
- Developed reproducible code and analytical materials; translated model outputs into operational decision scenarios
- Produced an open-access research paper and repository

Research Preparation

- Introduction to Agent-Based Modeling — Santa Fe Institute, Complexity Explorer (2026). Agent-based simulation, model conceptualisation, emergence, complex adaptive systems, collective behaviour, model verification, experimental design, BehaviorSpace experimentation, reproducible simulation. MOSAIC originated as the final project for this course and was subsequently expanded as an independent collaborative research project.
- Robotics Software Engineer Nanodegree — Udacity. Robot software engineering, localisation, path planning, search algorithms, mapping, autonomous navigation, robotics software concepts, and ROS-based development concepts.

Research Skills

Computational Research: Research problem formulation, experimental design, Agent-Based Modelling, computational simulation, statistical analysis, model verification and validation, reproducible research, scientific programming, quantitative interpretation, research documentation

Autonomous and Multi-Agent Systems: Multi-Agent Systems, mission-level coordination, distributed decision-making, agent communication, adaptive task allocation, coalition formation, market-based coordination, capability-based role allocation, autonomous cognitive systems, agent governance, human-autonomy teaming, human-in-the-loop supervision

Artificial Intelligence and Analytics: Machine learning, explainable AI, natural language processing, topological data analysis, behavioural segmentation, survival analysis, process mining, data mining, data engineering, decision-support systems, business analytics, operational analytics

Programming and Software Engineering: Python, R, SQL, NetLogo, object-oriented programming, database systems, Git/GitHub, software architecture, data pipelines, open-source research workflows

Analytical Methods and Technologies: XGBoost, logistic regression, random forests, Random Survival Forest, Kaplan-Meier estimation, Cox proportional hazards modelling, SHAP, UMAP, HDBSCAN, statistical model comparison, financial simulation

Robotics Foundations: Localisation, path planning, search algorithms, autonomous navigation, robot software concepts, ROS concepts

Business and Decision Analytics: Business analysis, decision analysis, requirements analysis, stakeholder analysis, business process modelling, process improvement, performance measurement, KPI design, operational analysis, scenario analysis, cost-benefit analysis, risk analysis, data-driven decision-making, decision-support systems

Economic Foundations and Market Processes: Economic fundamentals, microeconomic reasoning, price theory, marginal analysis, opportunity cost, market processes, economic calculation, entrepreneurship, competition and coordination, incentives, resource allocation, capital and interest, monetary theory, business cycle theory, institutional analysis, spontaneous order, methodological individualism

Project and Research Management: Project management, research project planning, scope management, risk management, stakeholder management, requirements management, resource planning, schedule management, project governance, project monitoring and control, agile project management, cross-functional coordination, research delivery

Professional Memberships

- Australian Computer Society — Associate Member, No. 4448962
- IEEE (Institute of Electrical and Electronics Engineers) — Member, No. 101391129

Open Research and Professional Profiles

ORCID: orcid.org/0009-0002-7194-9021

GitHub: github.com/ka-means

LinkedIn: linkedin.com/in/katherin-molina

Kaggle: kaggle.com/kathmolina

Zenodo: zenodo.org/records/19815868

Languages

Spanish
Native

English

Full Professional Proficiency

References

Professor Pablo Marshall Rivera

Professor Marcos Ernesto Sepúlveda Fernández

Additional References